

Tenth day of class - 9/25

Fun fact: The largest positive integer n for which $2^n - 7$ is a perfect square is $n = 15$. We have $2^{15} - 7 = 32761 = 161^2$.

You can prove this with elliptic curves! An elliptic curve has finitely many integral points (and Magma can provably find them, if it can compute the rank).

Write $n = 3k + r$ where $r \in \{0, 1, 2\}$ and set $x = 2^k$. Then $2^r x^3 - 7 = y^2$. If $r = 0$, this gives $y^2 = x^3 - 7$.

If $r = 1$, $y^2 = 2x^3 - 7$. You can multiply through by 4 and get $4y^2 = 8x^3 - 28$ and so you get an integral point on $Y^2 = X^3 - 28$.

If $r = 2$, $y^2 = 4x^3 - 7$. You can multiply through by 16 and get $16y^2 = 64x^3 - 112$ and so you get an integral point on $Y^2 = X^3 - 112$.

Last time: An introduction to group theory, and some discussion of stating results about elliptic curves in the language of group theory.

Example: Let $E : y^2 = x^3 - 16x + 16$. What is T ?

Claim: $T = \{(0 : 1 : 0)\}$.

Suppose not. Suppose that $P \in E(\mathbb{Q})$ has finite order and $P \neq (0 : 1 : 0)$. Then $\phi_3(P) \in E(\mathbb{F}_3) \cong \mathbb{Z}/7\mathbb{Z}$ but it cannot have order 1. Thus, $\phi_3(P)$ has order 7. Then $7\phi_3(P) = \phi_3(7P) = (0 : 1 : 0)$. We know that $7P$ has to have finite order (because P does), but the Nagell-Lutz theorem shows that the only way $7P$ can reduce to the point at infinity mod 3 is if it equals $(0 : 1 : 0)$. Thus, $7P = (0 : 1 : 0)$ and so P has order 7.

On the other hand, $E(\mathbb{F}_5) \cong \mathbb{Z}/8\mathbb{Z}$ and so $\phi_5(P)$ must have order that divides 7. But the only element of $E(\mathbb{F}_5)$ with order dividing 7 is $(0 : 1 : 0)$ and this implies that $\phi_5(P) = (0 : 1 : 0)$. But this contradicts the Nagell-Lutz theorem.

So there aren't any points of finite order other than $(0 : 1 : 0)$.

(The command `TorsionSubgroup(E)` in Magma will also return the finite abelian group T and an isomorphism between T and the *subgroup* of points of finite order in $E(\mathbb{Q})$.)

Theorem. If $E/\mathbb{Q}$ is an elliptic curve, then there is a non-negative integer r and a finite abelian group T so that

$$E(\mathbb{Q}) \cong \overbrace{\mathbb{Z} \times \mathbb{Z} \times \cdots \times \mathbb{Z}}^{r \text{ copies}} \times T.$$

Comments about r :

As mentioned last time, we don't know if there is an algorithm to compute r . There's much that we don't know about r . In 2006, Noam Elkies found an elliptic curve with $r \geq 28$ (we

think $r = 28$). Last November, Elkies and Klagsbrun found an elliptic curve with $r \geq 29$. (We don't know for sure that $r = 29$, but have good reason to suspect that this is the case.)

Some mathematicians think the value of r can be arbitrarily large.

Other mathematicians think there is an absolute limit on how large r can be. (There's a paper of Jennifer Park, Bjorn Poonen, John Voight, and Melanie Wood that predicts there are only finitely many elliptic curves $E/\mathbb{Q}$ with $r > 21$.)

There's also a conjecture that a "typical" elliptic curve should have $r = 0$ or $r = 1$ (with each occurring about 50 percent of the time.)

Extra problem on HW: Construct an elliptic curve with as high rank as you can. (Use Magma to compute the rank of your curve.) Please don't look up elliptic curves with high rank in databases and just use those curves.

You could just test random curves. (Like all $y^2 = x^3 + Ax + B$ with $|A|, |B| \leq 100$.) Or, you could try to find a systematic way to build curves with lots of rational points on them.

For students in MTH 346: Find a curve with rank ≥ 4 . If you just test random curves (like all $y^2 = x^3 + Ax + B$ with $|A|, |B| \leq 100$) you should find one of rank 4.

For students in MTH 646: Find a curve with rank ≥ 5 . You probably need a somewhat systematic method for coming up with curves that have lots of points.

People also search for curves of high rank with a given choice of T . Here are the records. In the table below, each of the recorded ranks is a lower bound.

T	Highest rank known
$\mathbb{Z}/1\mathbb{Z}$	29
$\mathbb{Z}/2\mathbb{Z}$	20
$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$	15
$\mathbb{Z}/3\mathbb{Z}$	15
$\mathbb{Z}/4\mathbb{Z}$	13
$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$	9
$\mathbb{Z}/5\mathbb{Z}$	9
$\mathbb{Z}/6\mathbb{Z}$	9
$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/6\mathbb{Z}$	6
$\mathbb{Z}/8\mathbb{Z}$	6
$\mathbb{Z}/7\mathbb{Z}$	6
$\mathbb{Z}/9\mathbb{Z}$	4
$\mathbb{Z}/10\mathbb{Z}$	4
$\mathbb{Z}/12\mathbb{Z}$	4
$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/8\mathbb{Z}$	3

Section 2.1 - Points of order 2 and 3

We say that P has order m on E if $P + P + \cdots + P = \mathcal{O}$. We will let $E[m]$ be the set of points on E of order dividing m . If K is a field, we write $E[m](K)$ to be the points of order dividing m that have coordinates in K .

Theorem. *Let C be a non-singular cubic curve of the form*

$$C : y^2 = f(x) = x^3 + ax^2 + bx + c.$$

- (a) *A point $P = (x, y) \neq \mathcal{O}$ has order 2 if and only if $y = 0$.*
- (b) *C has exactly four points of order 2 (over $\mathbb{C}$). These four points form a group which is isomorphic to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.*
- (c) *A point $P = (x, y) \neq \mathcal{O}$ has order three if and only if x is a root of the polynomial*

$$\psi_3(x) = 3x^4 + 4ax^3 + 6bx^2 + 12cx + (4ac - b^2).$$

- (d) *C has exactly nine points of order 3 (over $\mathbb{C}$). These nine points form a group which is isomorphic to $\mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$.*

Note: $y^2 = f(x)$ is non-singular provided that $f(x)$ has no repeated root. This occurs if and only if $f(x)$ and $f'(x)$ have no common roots.

What if $m = 2$? This means that $P + P = \mathcal{O}$. We compute $P + P$ by drawing the tangent line to the curve. The tangent line goes through the point at infinity if and only if this tangent line is vertical. This means that if we have $y^2 = x^3 + ax^2 + bx + c$, then we get a vertical tangent if and only if $y = 0$. This is clear geometrically.

Algebraically, we have that if $P = (x_0 : y_0 : z_0)$ is on $F(x, y, z) = y^2z - x^3 - ax^2z - bxz^2 - cz^3$, the tangent line to the curve is

$$F_x(x_0, y_0, z_0)x + F_y(x_0, y_0, z_0)y + F_z(x_0, y_0, z_0)z = 0.$$

We get $F_x = -3x^2 - 2axz - bz^2$, $F_y = 2yz$, and $F_z = y^2 - ax^2 - 2bxz - 3cz^2$. If the point at infinity $(0 : 1 : 0)$ is on this curve, then $F_y = 0$ and so $y = 0$ or $z = 0$. Thus, the tangent goes through the point at infinity if and only if $y = 0$ or the point is $(0 : 1 : 0)$. Thus, there are four points in $E[2](\mathbb{C})$, namely $(0 : 1 : 0)$ or $(r_i : 0 : 1)$, where r_i is a root of the cubic.

If $E[2](\mathbb{R})$ has size 4, then the picture looks like this. (Draw picture.)

There are three points $T_1 = (r_1, 0)$, $T_2 = (r_2, 0)$ and $T_3 = (r_3, 0)$. We have that $T_1 + T_1 = T_2 + T_2 = T_3 + T_3 = \mathcal{O}$. Also,

$$T_1 + T_2 = T_3, T_1 + T_3 = T_2, T_2 + T_3 = T_1.$$

The subgroup $\{\mathcal{O}, T_1, T_2, T_3\}$ is isomorphic to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

(If $E(\mathbb{R})$ has only one component, then $E(\mathbb{R})[2]$ has size 2.)

What about points of order 3?

Recall the duplication formula. The x -coordinate of $2(x, y)$ is

$$\frac{x^4 - 2bx^2 - 8cx + b^2 - 4ac}{4x^3 + 4ax^2 + 4bx + 4c} = \frac{f'(x)^2}{4f(x)} - a - 2x.$$

We have that $3P = \mathcal{O}$ if and only if $2P = -P$. We have that $2P = -P$ if and only if the x -coordinates of $2P$ and $-P$ are the same.

(Conversely, if the x -coordinates of $2P$ and $-P$ are the same, then either $2P = -P$ and so $3P = \mathcal{O}$, or $2P = P$ and $P = \mathcal{O}$.)

The equation is therefore $\frac{f'(x)^2}{4f(x)} - a - 2x = x$. This gives us $\frac{f'(x)^2}{4f(x)} = 3x + a = \frac{f''(x)}{2}$. Hence, we want $\psi_3(x) = 2f''(x)f(x) - f'(x)^2 = 0$. This is the polynomial given above. This proves (c).

Claim: $\psi_3(x)$ has four distinct complex roots.

Proof: We need to rule out the possibility that $\psi_3(x)$ has a repeated root. If $\psi_3(x) = (x - \alpha)^2 g(x)$, then $\psi_3'(x) = 2(x - \alpha)g(x) + (x - \alpha)^2 g'(x)$, and so α is also a root of $\psi_3'(x)$. So we want to show that $\psi_3(x)$ and $\psi_3'(x)$ have no roots in common.

We have that $\psi_3'(x) = 2f'''(x)f(x) + 2f''(x)f'(x) - 2f'(x)f''(x) = 2f'''(x)f(x) = 2 \cdot 6f(x) = 12f(x)$.

Thus, if $\psi_3(x) = 0$, then $P = (x, \sqrt{f(x)})$ is a point with $3P = \mathcal{O}$. However, if $f(x) = 0$, then $P = (x, 0)$ must have order 2. This is a contradiction. QED Claim

If $\beta_1, \dots, \beta_4$ are the roots of $\psi_3(x)$, then let $\gamma_i = \sqrt{f(\beta_i)}$. Then we get eight points of order 3, namely $(\beta_i, \pm\gamma_i)$. These, together with the point at infinity are the points on E with order 3.

Finally, there are only two abelian groups up to isomorphism with order 9, namely $\mathbb{Z}/9\mathbb{Z}$ and $\mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$. Since all the points in $E[3](\mathbb{C})$ have order dividing 3, this must be the latter.

Note 1: The points of order three are the points of inflection - the points where $\frac{d^2y}{dx^2}$ vanishes.

Note 2: It is *always the case* that precisely 3 of the 9 points in $E[3](\mathbb{C})$ are real.